

Syllabus

This year's course structure

Part I

Introduction to Programming with Python

In the first part, an introduction to the basic concepts of programming in Python is provided. Students will learn the Python syntax, data types, as well as how to implement loops, functions, and object classes in Python. Once these concepts are understood, we will learn how they can be used to solve problems.

Lectures

Welcome and Introduction (I)

Basics of Python syntax, variables, data types

Control Structures for Your Code (II)

String methods, comparisons, conditional statements, loops

Building Reusable Functions (III)

Functions, arguments, return values, scope, classes

Handling Data in more than one Dimension (IV)

Tuples, lists, sets, dictionaries

Errors and Debugging (V)

Exceptions, try-except blocks, debugging

Part II

Data Science with Python

In the second part, we will cover basic data science tools in Python referring to data manipulation, descriptive and explorative analysis as well as visualization. At the end, an outlook will be provided on the next steps in Python.

Lectures

Using Modules and Packages (VI)

Standard libraries, random numbers and how to use them

NumPy for Scientific Computing (VII)

Fast array operations with NumPy

Pandas and AI Craft (VIII)

Pandas for data manipulation and AI

Data Visualization (IX)

Matplotlib with AI based on hands-on examples

Part III

Programming Projects

In the third part, students take on a final project in Python where they apply their new knowledge in pairs on a topic of their choice. Each pair will present their results and get feedback at the end of the semester.

Lectures

Tooling, Git and Your Project (X)

Bring your own machine with the Session IX pre-work done (git and the GitHub CLI, uv, Zed with Mistral Vibe), learn git and GitHub, form pairs, and choose your project

Project Work I (XI)

Progress your pair project under assistance

Project Work II (XII)

Finalize your pair project with your partner; the first presentations take place here

Presentations and Discussion (XIII)

Present your pair's work and the learnings you have made

Checkpoints

Five in-class checkpoints, 12 points each, always the **first 40 minutes** of the session, individual work, uploaded to Moodle before the time is up. No make-ups: the point math absorbs one miss. Session II has a 15-minute ungraded practice run in the same format.

Checkpoint	Opens	Covers	AI
0 (practice, ungraded)	Session II, before the lab	Session I	no
1	Session III	Sessions I-II	no
2	Session V	Sessions III-IV	no
3	Session VI	Sessions I-V	no
4	Session VIII	Sessions VI-VII	allowed
5	Session X	Sessions VIII-IX	allowed